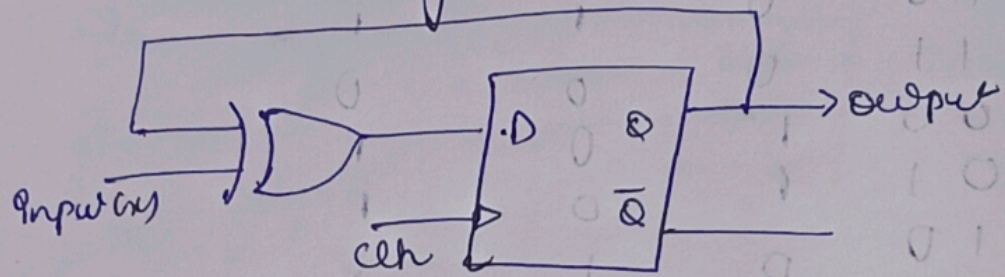


Sequential Assignment

1) Explain the behaviour of the given circuit.



from the above $D = x \oplus Q$ and $Q_{next} = Q$

Q	x	$D = x \oplus Q$	Q_{next}
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	1

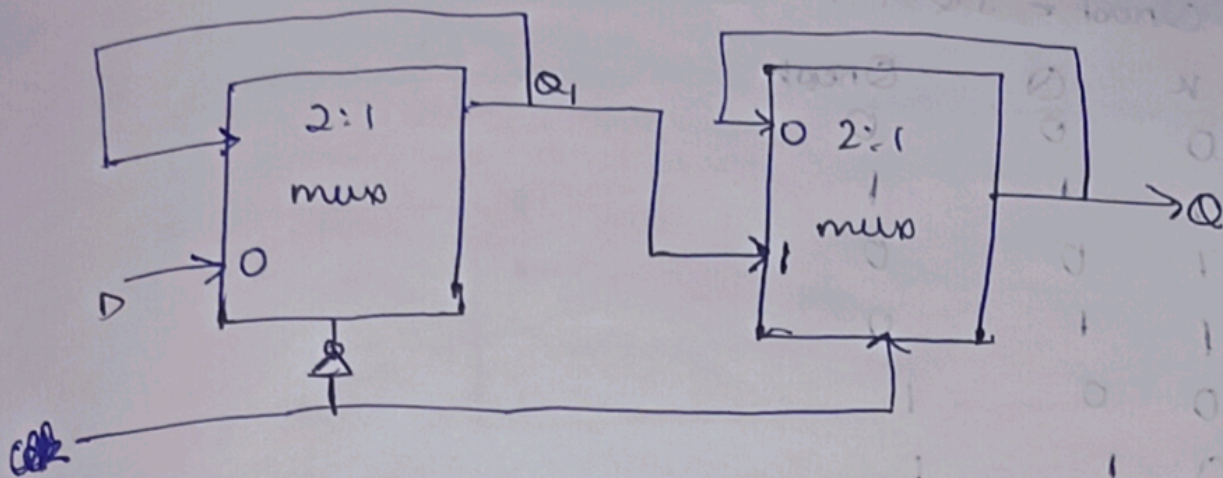
Comparing Input x and the next state Q_{next} we can say that

when $x = 0$ $Q_{next} = Q$ and

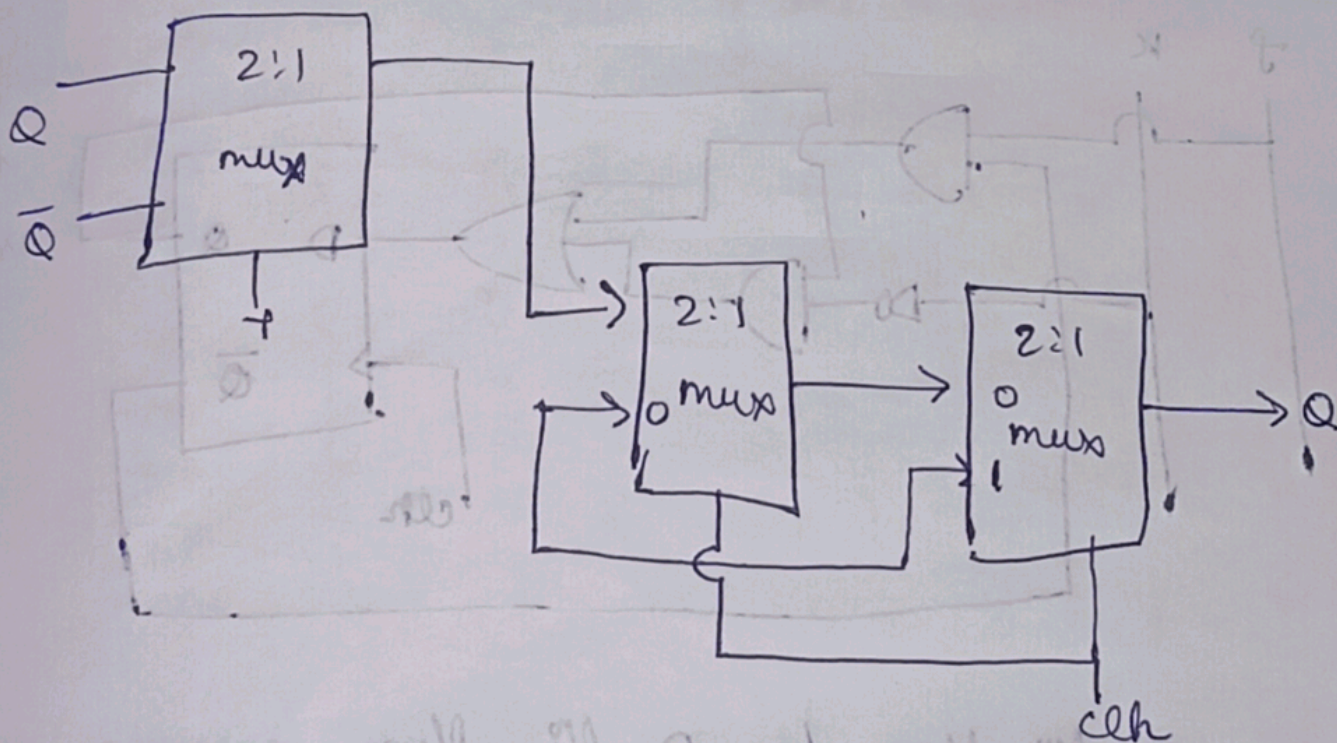
when $x = 1$ $Q_{next} = \bar{Q}$

$\therefore$ The behaviour of the given circuit is of flip flop

2) Design a posedge triggered D flip flop using 221 m only



3) Design a negedge triggered $\uparrow$ flip flop using 2:1 muxes only



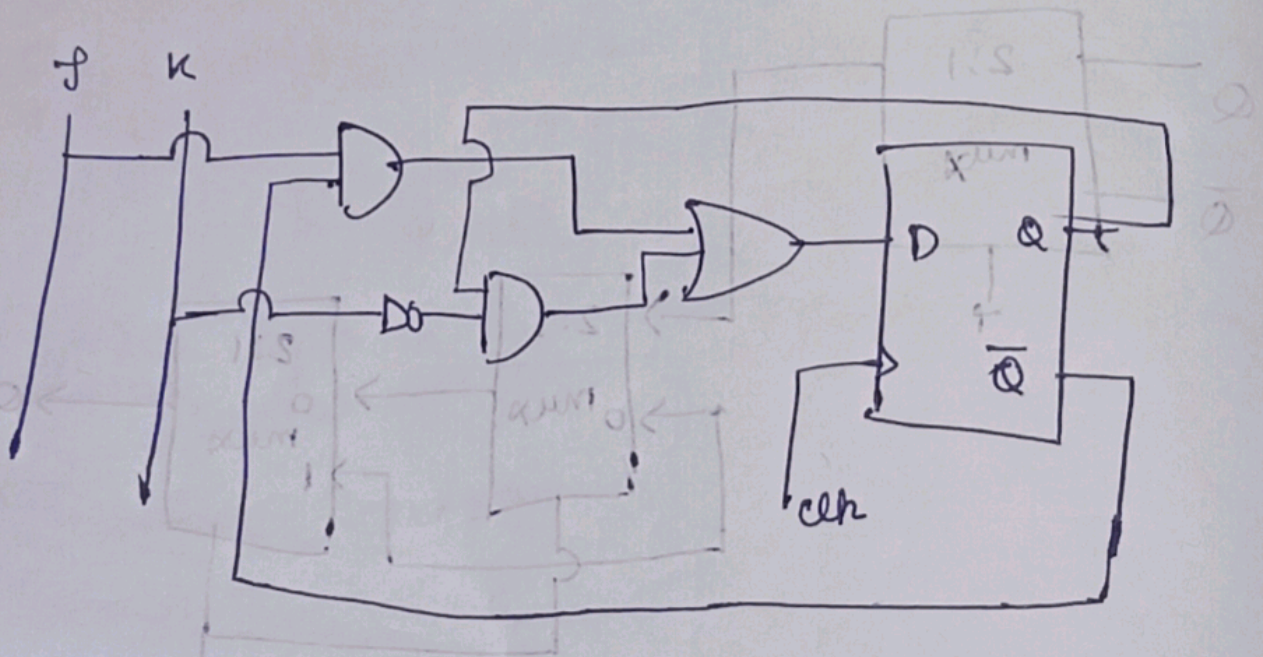
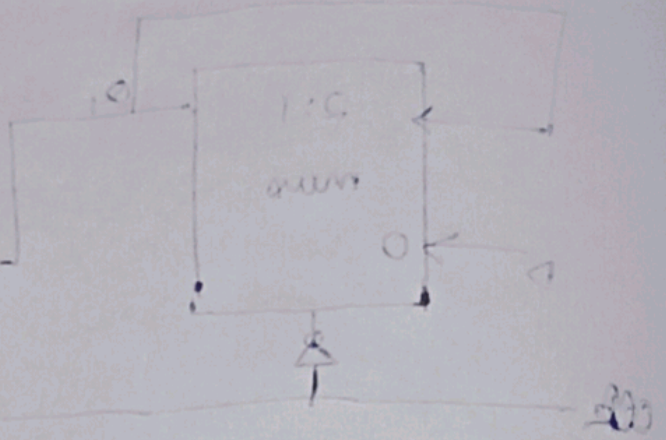
Convert by adding external gates.

A D flip flop to a J-K flip flop

	00	01	11	10
0	0	1	0	0
1	0	1	0	1

$$D = Q_{next} = S\bar{Q} + \bar{K}Q$$

S	K	Q	Q _{next}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

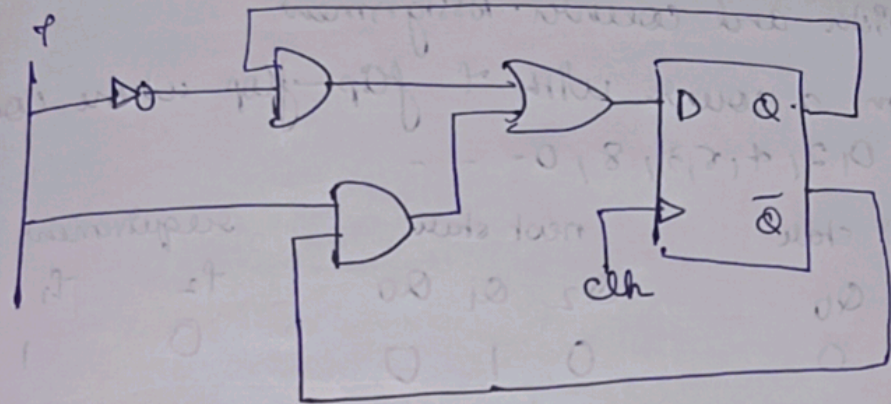


b) At T flip flop to D flip flop

T	Q	Q _{next}
0	0	0
0	1	1
1	0	1
1	1	0

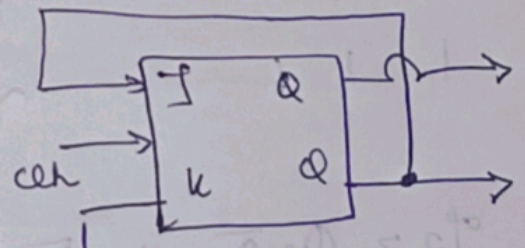
T	Q
0	0
0	1
1	0
1	1

$$Q_{next} = \bar{T}Q + T\bar{Q}$$



5) In a positive edge triggered JK flip flop we have $J = \bar{Q}$ and $K = 1$ as shown in the figure below. Assume the flip-flop was initially cleared and then clocked for 6 clock pulses. find the sequence of the Q output for those 6 clock cycles.

	J	K	Q	$\bar{Q}$
1st	1	1	1	0
2nd	0	1	0	1
3rd	1	1	1	0
4th	0	1	0	1
5th	1	1	1	0
6th	0	1	0	1



$\therefore$ the sequence $Q = 1, 0, 1, 0, 1, 0$

